Demo Exercise Sheet 10

Topology, Convex Sets, Unconstrained Optimization, and KKT Conditions

Operations Research

3rd July 2026

Roadmap for the Demo Exercise

1. Topology

Open/closed, boundedness, compactness, convexity.

2. Unconstrained opt.

Gradient, Hessian, stationary points, eigenvalue classification.

3. KKT

Conditions, constraint qualifications, global extrema under equality constraints.

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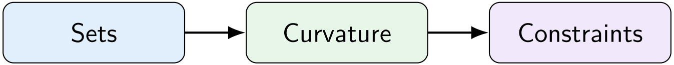

D10.1

D10.1: Given sets in $\mathbb{R}^2$

$$X_1 = \{x \in \mathbb{R}^2 \mid \|x\|_2 < 1\}, \qquad X_2 = \{x \in \mathbb{R}^2 \mid x_1 + x_2 \leq 1,\ x_1 \geq 0,\ x_2 \geq 0\}.$$

1. Determine whether X_1 and X_2 are open, closed, bounded, and compact.
2. Prove rigorously that X_2 is convex.

Why do we care?

Compactness of the domain + continuous f guarantees existence of optimum.
Convex Optimization methods require convex domain.

Quick Definitions to Use in the Proofs

Open

$X \subseteq \mathbb{R}^n$ is open if for every $x \in X$ there exists $\varepsilon > 0$ such that

$$B_\varepsilon(x) \subseteq X.$$

Closed

X is closed if it contains all of its limit points.
Equivalently, $\mathbb{R}^n \setminus X$ is open.

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Closed

X is closed if it contains all of its limit points. Equivalently, $\mathbb{R}^n \setminus X$ is open.

Bounded

X is bounded if $\exists M < \infty$ such that $\|x\| \leq M$ for all $x \in X$.

Compact in $\mathbb{R}^n$

By Heine-Borel:

$$X \text{ compact} \quad \Longleftrightarrow \quad X \text{ closed and bounded.}$$

D10.1(a): X_1 Is the Open Unit Ball

$$X_1 = \{x \in \mathbb{R}^2 \mid \|x\|_2 < 1\}.$$

- **Open:** If $x \in X_1$, choose

$$\varepsilon = \frac{1 - \|x\|_2}{2} > 0.$$

If $\|y - x\|_2 < \varepsilon$, then

$$\|y\|_2 \leq \|x\|_2 + \|y - x\|_2 < \|x\|_2 + \frac{1 - \|x\|_2}{2} < 1.$$

Thus $y \in X_1$.

D10.1(a): X_1 Is the Open Unit Ball

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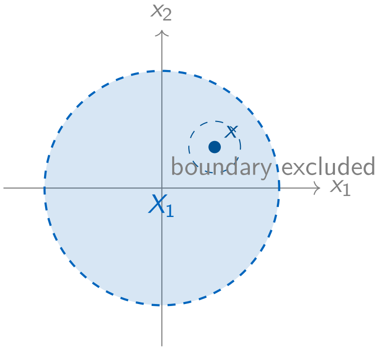
  Thus $y \in X_1$.
- **Bounded:** $\|x\|_2 < 1$ for all $x \in X_1$.



D10.1(a): Why X_1 Is Not Closed and Not Compact

Not closed

Consider the sequence

$$x^{(k)} = \left(1 - \frac{1}{k}, 0\right), \qquad k = 2, 3, \dots$$

Each $x^{(k)} \in X_1$, but

$$x^{(k)} \to (1, 0), \qquad \|(1, 0)\|_2 = 1,$$

so the limit point $(1, 0)$ is not contained in X_1.

Conclusion for X_1

X_1 is **open** and **bounded**, but **not closed**. Therefore, by Heine-Borel, X_1 is **not compact**.

D10.1(a): X_2 Is a Closed, Bounded Simplex

$$X_2 = \{x \in \mathbb{R}^2 \mid x_1 + x_2 \leq 1,\ x_1 \geq 0,\ x_2 \geq 0\}.$$

- **Closed:** It is the intersection of three closed halfspaces:

$$\{x : x_1 + x_2 \leq 1\} \cap \{x : x_1 \geq 0\} \cap \{x : x_2 \geq 0\}.$$

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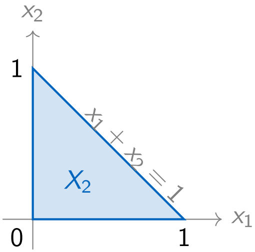


- **Closed:** It is the intersection of three closed halfspaces:

$$\{x : x_1 + x_2 \le 1\} \cap \{x : x_1 \ge 0\} \cap \{x : x_2 \ge 0\}.$$

- **Bounded:** From $x_1, x_2 \ge 0$ and $x_1 + x_2 \le 1$,

$$0 \le x_1 \le 1, \qquad 0 \le x_2 \le 1.$$

  Hence $X_2 \subseteq [0,1]^2$.

Conclusion: X_2 is closed and bounded, hence compact by Heine-Borel. It is not open because boundary points such as $(0,0)$ have no ball contained in X_2.

D10.1(a): Summary Table

Set	Open?	Closed?	Bounded?	Compact?
$X_1 = \{\|x\|_2 < 1\}$	Yes	No	Yes	No
$X_2 = \{x_1 + x_2 \leq 1,\ x_1, x_2 \geq 0\}$	No	Yes	Yes	Yes

Common mistakes to avoid

- "Strict inequality" often indicates openness, but prove it with an ε-ball or continuity argument.
- "Non-strict inequality" often indicates closedness, but only after recognizing it as a preimage of a closed set or a closed halfspace.
- Compactness in $\mathbb{R}^n$ is exactly closed plus bounded.

D10.1(b): Convexity of X_2

Definition

A set $C \subseteq \mathbb{R}^n$ is convex if for all $x, y \in C$ and all $\lambda \in [0, 1]$,

$$z = \lambda x + (1 - \lambda) y \in C.$$

D10.1(b): Convexity of X_2

Definition

A set $C \subseteq \mathbb{R}^n$ is convex if for all $x, y \in C$ and all $\lambda \in [0, 1]$,

$$z = \lambda x + (1 - \lambda)y \in C.$$

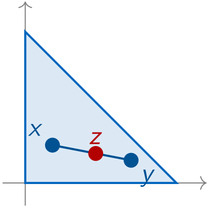
Let $x, y \in X_2$ and $\lambda \in [0, 1]$. Define $z = \lambda x + (1 - \lambda)y$.

$$z_i = \lambda x_i + (1 - \lambda)y_i \geq 0, \qquad i = 1, 2,$$

because all terms are nonnegative.

$$\begin{aligned} z_1 + z_2 &= \lambda(x_1 + x_2) + (1 - \lambda)(y_1 + y_2) \\ &\leq \lambda \cdot 1 + (1 - \lambda) \cdot 1 = 1. \end{aligned}$$

Thus z satisfies all constraints, so $z \in X_2$.



The line segment between any two feasible points stays feasible.

D10.2: Theory Recap

First-order necessary condition

If f is differentiable and x^* is a local extremum in the interior of the domain, then

$$\nabla f(x^*) = 0.$$

Such points are called **stationary points**.

Second-order classification for twice differentiable f

At a stationary point x^*:

- $H_f(x^*) \succ 0 \Rightarrow$ strict local minimum.
- $H_f(x^*) \prec 0 \Rightarrow$ strict local maximum.
- $H_f(x^*)$ indefinite $\Rightarrow$ saddle point.
- Semidefinite Hessian $\Rightarrow$ test is inconclusive.

D10.2

D10.2: Function

$$f : \mathbb{R}^2 \to \mathbb{R}, \qquad f(x, y) = x^2 y + y^3 - 3y.$$

1. Compute $\nabla f(x, y)$ and $H_f(x, y)$.
2. Determine all stationary points.
3. Classify each stationary point using the eigenvalues of the Hessian.

Structure of the solution

Derivatives $\to$ solve $\nabla f = 0$ $\to$ Hessian eigenvalues at each candidate.

D10.2(a): Gradient and Hessian

$$f(x, y) = x^2y + y^3 - 3y.$$

Gradient

$$\frac{\partial f}{\partial x} = 2xy, \qquad \frac{\partial f}{\partial y} = x^2 + 3y^2 - 3.$$

Therefore

$$\nabla f(x, y) = \begin{pmatrix} 2xy \\ x^2 + 3y^2 - 3 \end{pmatrix}.$$

D10.2(a): Gradient and Hessian

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Therefore

$$\nabla f(x, y) = \begin{pmatrix} 2xy \\ x^2 + 3y^2 - 3 \end{pmatrix}.$$

Hessian

$$H_f(x, y) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix} = \begin{pmatrix} 2y & 2x \\ 2x & 6y \end{pmatrix}.$$

D10.2(b): Stationary Points

Set $\nabla f(x, y) = 0$:

$$2xy = 0, \qquad x^2 + 3y^2 - 3 = 0.$$

The first equation gives $x = 0$ or $y = 0$.

Case 1: $x = 0$

$$3y^2 - 3 = 0 \quad \Longrightarrow \quad y = \pm 1.$$

Candidates:

$$(0, 1), \qquad (0, -1).$$

D10.2(b): Stationary Points

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Case 1: $x = 0$

$$3y^2 - 3 = 0 \implies y = \pm 1.$$

Candidates:

$$(0, 1), \qquad (0, -1).$$

Case 2: $y = 0$

$$x^2 - 3 = 0 \implies x = \pm\sqrt{3}.$$

Candidates:

$$(\sqrt{3}, 0), \qquad (-\sqrt{3}, 0).$$

D10.2(b): Stationary Points

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Case 2: $y = 0$

$$x^2 - 3 = 0 \quad \implies \quad x = \pm\sqrt{3}.$$

Candidates:

$$(\sqrt{3}, 0), \qquad (-\sqrt{3}, 0).$$

Result

There are exactly four stationary points: $(0, 1)$, $(0, -1)$, $(\sqrt{3}, 0)$, $(-\sqrt{3}, 0)$.

D10.2(c): Classification - Diagonal Cases

$$H_f(x,y) = \begin{pmatrix} 2y & 2x \\ 2x & 6y \end{pmatrix}.$$

At $(0,1)$

$$H_f(0,1) = \begin{pmatrix} 2 & 0 \\ 0 & 6 \end{pmatrix}.$$

Eigenvalues: 2 and 6.

$H_f(0,1) \succ 0 \quad \Longrightarrow \quad$ **strict local minimum.**

At $(0,-1)$

$$H_f(0,-1) = \begin{pmatrix} -2 & 0 \\ 0 & -6 \end{pmatrix}.$$

Eigenvalues: -2 and -6.

$H_f(0,-1) \prec 0 \quad \Longrightarrow \quad$ **strict local maximum.**

D10.2(c): Classification - Saddle Cases

At $(\sqrt{3}, 0)$

$$H_f(\sqrt{3}, 0) = \begin{pmatrix} 0 & 2\sqrt{3} \\ 2\sqrt{3} & 0 \end{pmatrix}.$$

$$\det(H - \lambda I) = \det \begin{pmatrix} -\lambda & 2\sqrt{3} \\ 2\sqrt{3} & -\lambda \end{pmatrix} = \lambda^2 - 12.$$

Eigenvalues: $\lambda = \pm 2\sqrt{3}$.

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Eigenvalues: $\lambda = \pm 2\sqrt{3}$.

At $(-\sqrt{3}, 0)$

$$H_f(-\sqrt{3}, 0) = \begin{pmatrix} 0 & -2\sqrt{3} \\ -2\sqrt{3} & 0 \end{pmatrix}.$$

$$\det(H - \lambda I) = \det \begin{pmatrix} -\lambda & -2\sqrt{3} \\ -2\sqrt{3} & -\lambda \end{pmatrix} = \lambda^2 - 12.$$

Eigenvalues: $\lambda = \pm 2\sqrt{3}$.

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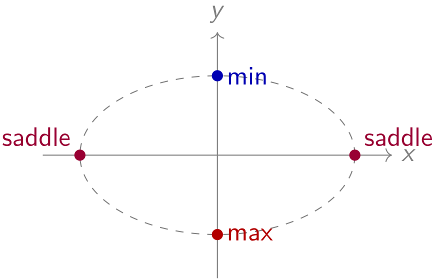
Eigenvalues: $\lambda = \pm 2\sqrt{3}$.

Conclusion

In both cases the Hessian has one positive and one negative eigenvalue, so it is indefinite. Both points are **saddle points**.

D10.2: Classification Summary

Stationary point	Eigenvalues of H_f	Type
$(0, 1)$	$2, 6$	Local min
$(0, -1)$	$-2, -6$	Local max
$(\sqrt{3}, 0)$	$-2\sqrt{3}, 2\sqrt{3}$	Saddle
$(-\sqrt{3}, 0)$	$-2\sqrt{3}, 2\sqrt{3}$	Saddle



stationary equation: $x^2 + 3y^2 = 3$

D10.3 Recap: What KKT Can Tell Us

Convex problem + Slater

Slaters Conditions:

$$\min f(x) \quad \text{s.t.} \quad g_i(x) \leq 0,\; h_i(x) = 0,$$

with f, g_i convex and h_i affine.
There exists a strictly feasible point x:
$h_i(x) = 0, g_i(x) < 0$

$$\text{Slater} \Longrightarrow \text{KKT} \iff \text{global optimum.}$$

General problem + LICQ

At a feasible point x, LICQ holds if

$$\{\nabla g_i(x^*) \mid g_i(x^*) = 0\} \cup \{\nabla h_j(x^*) \mid j = 1, \dots, p\}.$$

are independent.

$$\text{local optimum} \Longrightarrow \text{KKT point.}$$

KKT gives candidates; compare values for global extrema.

For one equality constraint

If the only constraint is $h(x) = 0$, then regularity condition reduces to

$$\nabla h(x^*) \neq 0.$$

D10.3: KKT

Let $f : \mathbb{R}^n \to \mathbb{R}$, $g : \mathbb{R}^n \to \mathbb{R}^m$, $h : \mathbb{R}^n \to \mathbb{R}^p$ and consider

$$\min_x f(x) \quad \text{s.t.} \quad g(x) \leq 0, \qquad h(x) = 0.$$

(a) **(4P)** State the KKT conditions.

(b) **(1P)** What changes for a maximization problem?

$$f(x, y) = \exp(2x^2 + 4y^2 + 24y), \qquad h(x, y) = x^2 + (y + 3)^2 - 1.$$

$$(P) \qquad \min f(x, y) \quad \text{s.t.} \quad h(x, y) = 0.$$

(c) Show that f is convex.

(d) Discuss Slater's condition for (P).

(e) Compute ∇f and ∇h.

(f) Show $\nabla h \neq 0$ at all feasible points.

(g) Explain how to determine global extrema using the KKT points $z_1, \ldots, z_4$.

D10.3(a): KKT Conditions

Assumptions for necessity

f, g_i, h_j are C^1, x^* is a local minimizer, and a constraint qualification holds.

$$\min f(x) \quad \text{s.t.} \quad g_i(x) \leq 0,\ h_j(x) = 0.$$

Then there exist multipliers λ_i and μ_j such that

$$\begin{aligned}
\textbf{Stationarity:} \quad & \nabla f(x^*) + \sum_i \lambda_i \nabla g_i(x^*) + \sum_j \mu_j \nabla h_j(x^*) = 0, \\
\textbf{Primal feasibility:} \quad & g_i(x^*) \leq 0, \qquad h_j(x^*) = 0, \\
\textbf{Dual feasibility:} \quad & \lambda_i \geq 0, \\
\textbf{Complementary slackness:} \quad & \lambda_i g_i(x^*) = 0.
\end{aligned}$$

Remember

In convex problems with Slater, these conditions also characterize global optima.

D10.3(b): Maximization

Convert to minimization

$$\max f(x) \quad \Longleftrightarrow \quad \min -f(x).$$

For constraints $g_i(x) \leq 0$ and $h_j(x) = 0$, stationarity becomes

$$-\nabla f(x^*) + \sum_i \lambda_i \nabla g_i(x^*) + \sum_j \mu_j \nabla h_j(x^*) = 0, \qquad \lambda_i \geq 0.$$

Unchanged

Primal feasibility, dual feasibility, and complementary slackness stay the same.

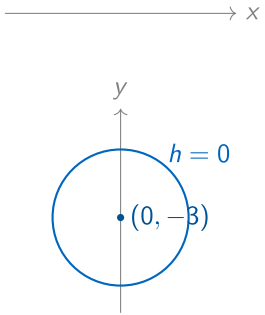
D10.3(c)–(g): Geometry of (P)

$$f(x, y) = \exp(2x^2 + 4y^2 + 24y)$$
$$h(x, y) = x^2 + (y + 3)^2 - 1.$$

The feasible set is

$$x^2 + (y + 3)^2 = 1,$$

a circle centered at $(0, -3)$ with radius 1.



Important

The feasible set is compact, but not convex.

D10.3(c): Convexity of the Objective

Write

$$f(x,y) = f_1(f_2(x,y)), \qquad f_1(t) = e^t, \qquad f_2(x,y) = 2x^2 + 4y^2 + 24y.$$

f_1

$$f_1'(t) = f_1''(t) = e^t > 0.$$

So f_1 is convex and strictly increasing.

f_2

$$H_{f_2} = \begin{pmatrix} 4 & 0 \\ 0 & 8 \end{pmatrix} \succ 0.$$

So f_2 is convex.

Conclusion

By the composition rule, $f = f_1 \circ f_2$ is convex.

D10.3(d): Slater's Condition

In (P),

$$h(x, y) = x^2 + (y + 3)^2 - 1 = 0$$

is a nonlinear equality constraint.

Conclusion

Standard Slater is not applicable. We therefore use LICQ.

D10.3(e)–(f): Gradients and Regularity

Gradients

$$\nabla f(x,y) = f(x,y)\begin{pmatrix} 4x \\ 8(y+3) \end{pmatrix}, \qquad \nabla h(x,y) = \begin{pmatrix} 2x \\ 2(y+3) \end{pmatrix}.$$

Check LICQ

$$\nabla h(x,y) = 0 \quad \iff \quad (x,y) = (0,-3).$$

But

$$h(0,-3) = -1 \neq 0,$$

so this point is not feasible.

Conclusion

$\nabla h \neq 0$ at every feasible point. Hence every feasible point is regular, and KKT is necessary for local extrema.

D10.3(g): Strategy for Global Extrema

1. **Existence:** compact feasible set + continuous f.
2. **KKT necessity:** $\nabla h \neq 0$ on the feasible set.
3. **Candidates:** solve

$$\nabla f(x, y) + \mu \nabla h(x, y) = 0, \qquad h(x, y) = 0.$$

4. **Decision:** evaluate f at all KKT points $z_1, \ldots, z_4$.

Key idea

Here KKT gives candidates, not automatic global optima. Compare the objective values.

Optional: D10.3(g) - KKT Candidate Calculation I

Let

$$u = y + 3, \qquad h(x, y) = x^2 + u^2 - 1.$$

Then

$$\nabla f(x, y) = f(x, y) \begin{pmatrix} 4x \\ 8u \end{pmatrix}, \qquad \nabla h(x, y) = \begin{pmatrix} 2x \\ 2u \end{pmatrix}.$$

For the equality-constrained problem, KKT gives

$$\nabla f(x, y) + \mu \nabla h(x, y) = 0, \qquad x^2 + u^2 = 1.$$

Thus

$$x(4f + 2\mu) = 0, \qquad u(8f + 2\mu) = 0, \qquad x^2 + u^2 = 1.$$

Optional: D10.3(g) - KKT Candidate Calculation II

From

$$x(4f + 2\mu) = 0, \qquad u(8f + 2\mu) = 0, \qquad x^2 + u^2 = 1,$$

we distinguish cases.

Case 1: $u = 0$

$$x^2 = 1 \quad \Rightarrow \quad x = \pm 1.$$

Thus

$$(1, -3), \quad (-1, -3).$$

Case 2: $x = 0$

$$u^2 = 1 \quad \Rightarrow \quad u = \pm 1.$$

Thus

$$(0, -2), \quad (0, -4).$$

If $x \neq 0$ and $u \neq 0$, then

$$\mu = -2f \quad \text{and} \quad \mu = -4f,$$

which is impossible since $f > 0$.

Optional: D10.3(g) - Global Minima and Maxima

The four KKT candidates are

$$(1, -3), \quad (-1, -3), \quad (0, -2), \quad (0, -4).$$

Their objective values are

$$f(1, -3) = f(-1, -3) = e^{-34},$$

$$f(0, -2) = f(0, -4) = e^{-32}.$$

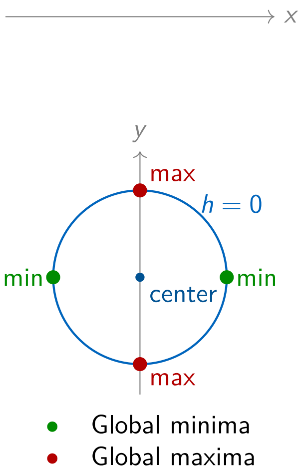
Since

$$e^{-34} < e^{-32},$$

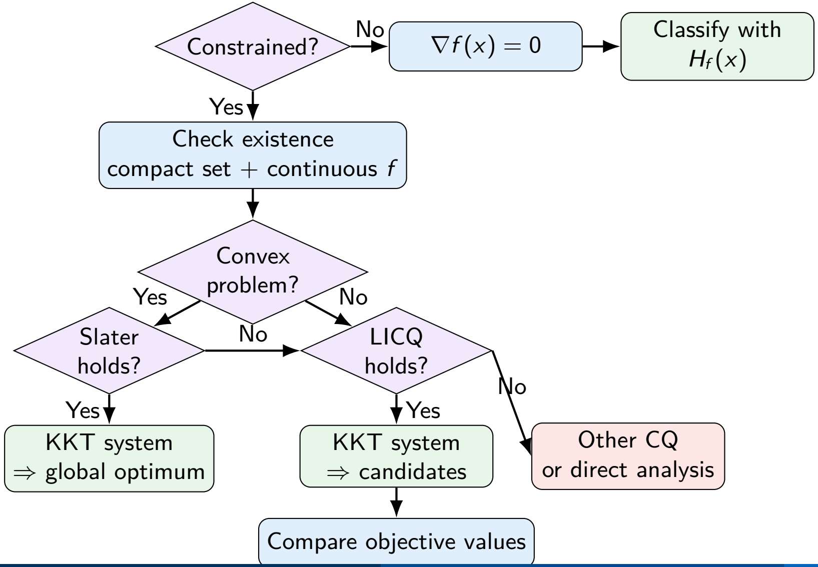
we obtain:

Global extrema

$$\text{Minima:} \quad (1, -3),\ (-1, -3),$$
$$\text{Maxima:} \quad (0, -2),\ (0, -4).$$



Overview



Key roles

Compactness
guarantees existence.

CQ / LICQ
justifies KKT necessity.

Regularity Condition (Slater, LICQ, ...)
makes KKT sufficient for global optimality.

For this exercise

Constrained, compact feasible set, Slater not applicable, LICQ holds.
So: KKT gives candidates, then compare values.

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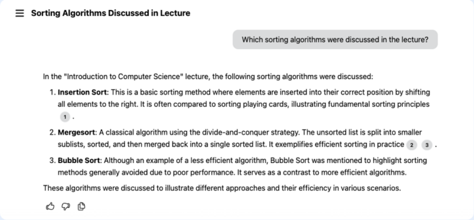

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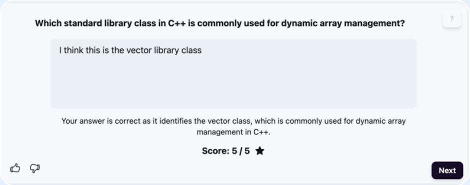

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